DSE6111

Predictive Modeling
Session # 5
Variable Selection & Regularization

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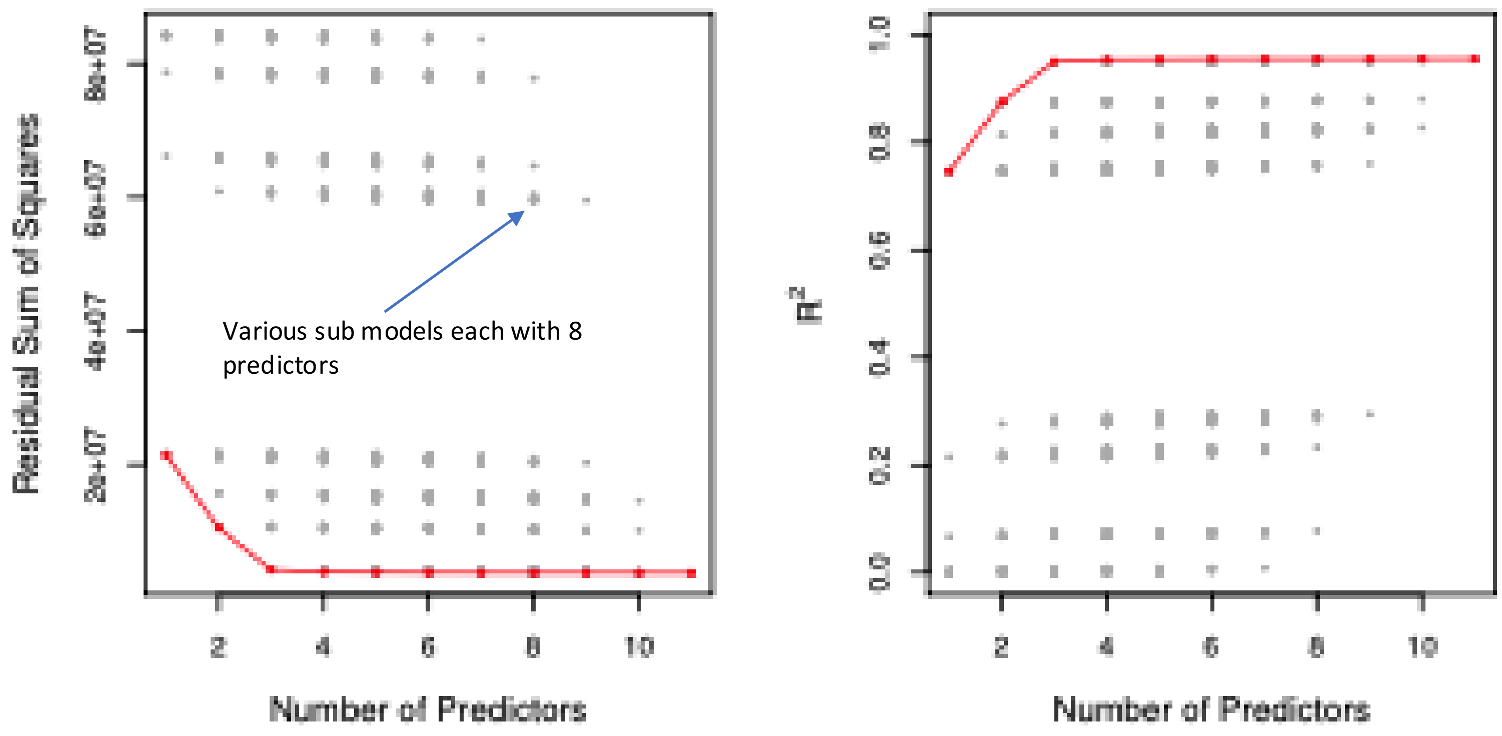
Best Subset Selection

- Recall the linear model: $Y = \beta_0 + \beta_1 * x_1 + \beta_p * x_p + \epsilon_i$
- The purpose for constructing the equation is to determine the magnitude by which the value of the independent variables must be altered to obtain a specified value of the response variable. The regression equation is viewed as a response function, with ***Y*** as the response variable and for this purpose it is desired that the coefficients **β** of the variable in the equation be measured accurately.
- But the problem is how to decide which variables to include or exclude, and how to help ensure you weren't leaving out any important variables.
- For cases when there are many potential predictor variables, a set of procedures that does not involve computing of all possible equation can be used. These procedures can help identify candidate variables by addition or deletion from the equation, one at a time and involves examining only a subset of all possible equations.
- With ***q*** variables these procedures will involve evaluation of at most ***q+1*** equations (Stepwise) as contrasted with the evaluation of 2^q equations (Best subsets) necessary
- The procedure can therefore be classified into two broad categories;
  - Stepwise regression and Best Subsets regression.
  - These automatic procedures can help when you have many independent variables, and you need assistance in the investigative stages of the variable selection process. You could specify many models with different combinations of independent variables, or you can have your statistical software do this for you.
  - These procedures are especially useful when theory and experience provide only a vague sense of which variables you should include in the model. However, if theory and expertise are strong guides, it's generally better to follow them than to use an automated procedure.
- Additionally, if you use one of these procedures, you should consider it as only the first step of the model selection process.

Best Subset Selection

- Best subsets regression is also known as “all possible regressions” and “all possible models”.
- Best subsets regression fits all possible models based on the independent variables that you specify.
- If there are 10 independent variables, it fits 1024 models. However, if there are 20 variables, it fits 1,048,576 models!
- Best subsets regression fits 2^P models, where P is the number of predictors in the dataset.
- To perform best subset selection, we fit a separate least squares regression for each possible combination of the ***p*** predictors.
- The problem of selecting the best model from among the 2^p possibilities considered by best subset selection is not trivial
  - Let M_0 denote the null model, which contains no predictors. This model simply predicts the sample mean for each observation.
    - For k = 1,2,...p:
      - Fit all $\binom{p}{k}$ models that contain exactly k predictors.
      - Pick the best among these $\binom{p}{k}$ models, and call it M_k. Here best is defined as having the smallest SSR (Sum of Square Residuals), or equivalently largest R^2.
      - Select a single best model from among $M_0, \ldots, M_P$ using cross-validated prediction error, C_p , (AIC), BIC or adjusted R^2.

Best Subset Selection: Example



- For each possible model containing a subset of the ten predictors in the Credit data set, the SSR and R^2 are displayed.
- The red frontier tracks the best model for a given number of predictors, according to SSR and R^2
- Though the data set contains only ten predictors, the x-axis ranges from 1 to 11, since one of the variables is categorical and takes on three values, leading to the creation of two dummy variables

Best subset selection limitation

- For computational reasons, best subset selection cannot be applied with very large p. *Why not?*
- Best subset selection may also suffer from statistical problems when p is large: larger the search space, the higher the chance of finding models that look good on the training data, even though they might not have any predictive power on future data.
- Thus an enormous search space can lead to *overfitting* and high variance of the coefficient estimates.
- For both of these reasons, *stepwise* methods, which explore a far more restricted set of models, are attractive alternatives to best subset selection.

Forward Selection Procedure

- Forward stepwise selection begins with a model containing no predictors, and then adds predictors to the model, one-at-a-time, until all the predictors are in the model.
- The first variable included in the equation is the one which has the highest simple correlation with the response variable ***Y***.
- If the regression coefficient of this variable is significantly different from zero, it is retained in the equation, and a search for second variable is made.
- The variable that enters the equation as the second variable is the one which has the highest correlation with ***Y***, after ***Y*** has been adjusted for the effect of the first variable i.e., the variable with the highest simple correlation coefficient with the residuals from Step 1.
- The significance of the regression coefficient of the second variable is then tested, If the regression coefficient is significant, a search for a third variable is made in the same way. The procedure is terminated when the last variable entering the equation is not significant

# Variables	Best subset	Forward stepwise
One	rating	rating
Two	rating, income	rating, income
Three	rating, income, student	rating, income, student
Four	cards, income, student, limit	rating, income, student, limit

The first four selected models for best subset selection and forward stepwise selection on the Credit data set. The first three models are identical but the fourth models differ.

- Computational advantage over best subset selection is clear.
- It is not guaranteed to find the best possible model out of all 2^p models containing subsets of the p predictors.

Backward Selection Procedure

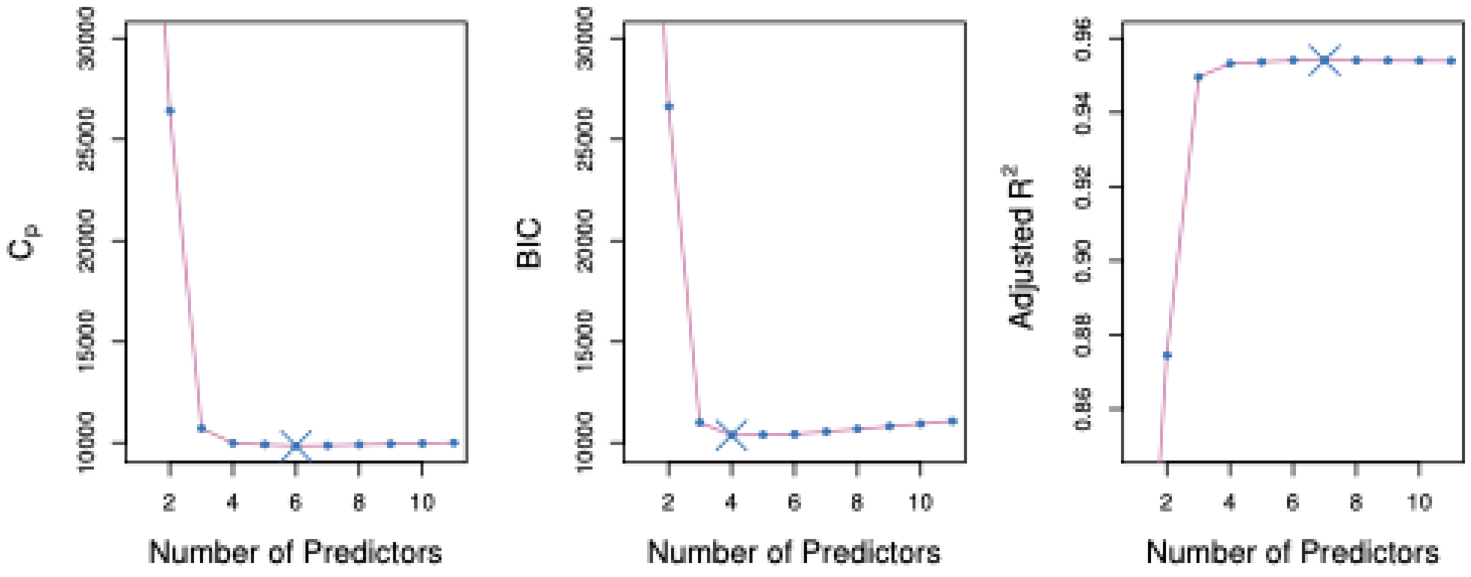
- The backward elimination procedure starts with the full equation and successively drops one variable at a time.
- The variables are dropped based on their contribution to the reduction of SSE (Sum of Squares Errors).
- The first variable deleted is the one with the smallest contribution to the reduction of SSE. This is equivalent to deleting the variable which has the smallest t-Test in the equation.
- If all the t-Tests are significant, the full set of variables are retained in the equation.
- Assuming that there are one or more variables that are not significant, the procedure operates by dropping the variable with the smallest not significant t-test.
- The equation with the remaining ***q-1*** variables is then fitted and the t-tests for the regression coefficients are examined, The procedure is terminated when all the t-tests are significant.
- Backward selection requires that the number of samples n is larger than the number of variables p (so that the full model can be fit).
- In contrast, forward stepwise can be used even when $n < p$, and so is the only viable subset method when p is very large.

Stepwise Selection Procedure

- The stepwise method is essentially a forward selection procedure but with the added proviso that at each stage the possibility of deleting a variable, as in backward elimination, is considered.
- In this procedure a variable that entered in the earlier stages of selection maybe eliminated at later stages.
- The calculation made for inclusion and deletion of variables are the same as FS & BE procedures.

Criteria for Evaluating Models & Equations

- We have observed that the training set MSE is generally an under- estimate of the test MSE. (Recall that MSE = SSR/n)
- This is because when we fit a model to the training data using least squares, we specifically estimate the regression coefficients such that the training SSR (but not the test SSR) is as small as possible.
- In particular, the training error will decrease as more variables are included in the model, but the test error may not. Therefore, training set SSR and training set R^2 cannot be used to select from among a set of models with different numbers of variables.
- To judge the adequacy of various fitted equations we need a criterion.
- C_p, AIC, BIC and Adjusted R^2 are some of the most popular methods



Mallows' C_p

- Mallows' C_p helps you choose between multiple regression models by striking a balance between precision and bias. You want to include enough independent variables to eliminate bias but not too many to reduce precision.
- This balance changes with the number of independent variables in the model. Mallows' C_p compares the precision and bias of the full model to models with a subset of the predictors. Typically, you want Mallows' C_p to be small and close to the number of variables in the model plus the constant.
- For a fitted least squares model containing ***d*** predictors, the C_p estimate of test MSE is computed using the equation

$$C_p = \frac{1}{n}(\text{SSR} + 2\text{d}\,\widehat{\sigma^2})$$

  where where ***d*** is the total # of predictors used and $\widehat{\sigma^2}$ is an estimate of the variance of the error ε associated with each response measurement (MSE).

- Typically, $\widehat{\sigma^2}$ is estimated using the full model containing all predictors. Essentially, the C_p statistic adds a penalty of ***2d*** $\boldsymbol{\widehat{\sigma^2}}$ to the training SSR to adjust for the fact that the training error tends to underestimate the test error.
- The penalty increases as the number of predictors in the model increases; this is intended to adjust for the corresponding decrease in training SSR. Therefore, the $\mathbf{C_p}$ statistic tends to take on a small value for models with a low-test error, so when determining which of a set of models is best, we choose the model with the lowest C_p value.
- If regression is purely for prediction, all the models with C_p < p, predict about equally well, in which case there's no reason to get carried away about exactly which model is the "best".

AIC & BIC

$$C_p = \frac{1}{n}(\text{SSR} + 2d\,\widehat{\sigma^2})$$

- The *Akaike information criterion* (*AIC*) and the *Bayesian information criterion* (*BIC*), are currently the most used model selection criteria beyond classical hypothesis tests. Both are members of a more general family of *penalized* model-fit statistics, applicable to regression models fit by maximum likelihood. It is a way to balance bias and variance or accuracy (fit) and simplicity (parsimony).
- It is an estimate of the or KL divergence between candidate models and true model, on a log-scale. The models that are closer (have a smaller distance) to the truth are better and we can compare how close two models are to the truth, picking the one that has a smaller distance (smaller AIC) as better, without the train & test split
- In the context of linear model AIC is like C_p

$$AIC = -2\log \mathrm{L} + 2\mathrm{d}$$

- The advantage of using the log-likelihood rather than SSE is that we have an AIC for linear models, probit, logit, poisson, etc.
- Like C_p, the BIC tends to take on a small value for a model with a low-test error, and so generally we select the model that has the lowest BIC value.

$$BIC = -2\log \mathrm{L} + \mathrm{d}\ln(\mathrm{n})$$

  - AIC & BIC can be generalized to broader class of models as they do not rely explicitly on residual based variance $\widehat{\sigma^2}$ used by C_p.
  - BIC is preferred for models where n < d

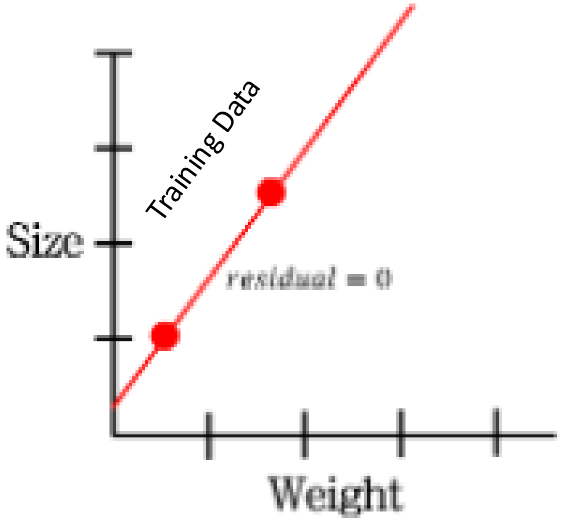
Adjusted R^2

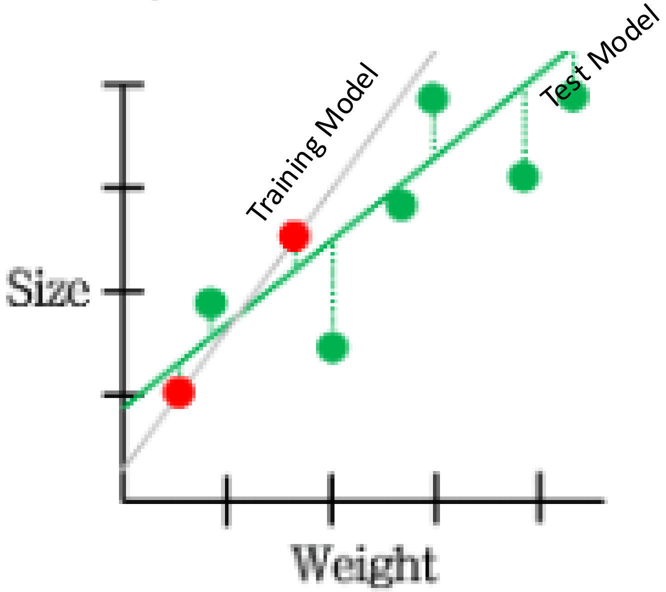
- The adjusted R^2 statistic is another popular approach for selecting among a set of models that contain different numbers of variables.
- The usual R^2 is defined as $1 - \frac{SSR}{SST}$ where SST = $\sum(y_i - \bar{y})^2$ is the total sum of squares for the response.
- Since SSR always decreases as more variables are added to the model, the R^2 always increases as more variables are added. For a least squares model with ***d*** variables, the adjusted R^2 statistic is calculated as

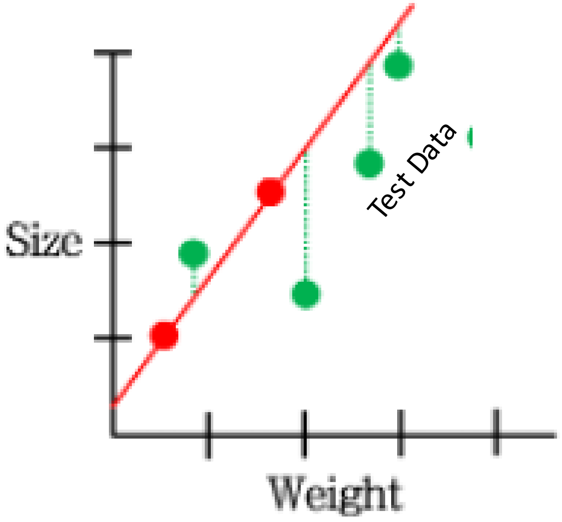
$$1 - \frac{SSR/(n-d-1)}{SST\ (n-1)}$$

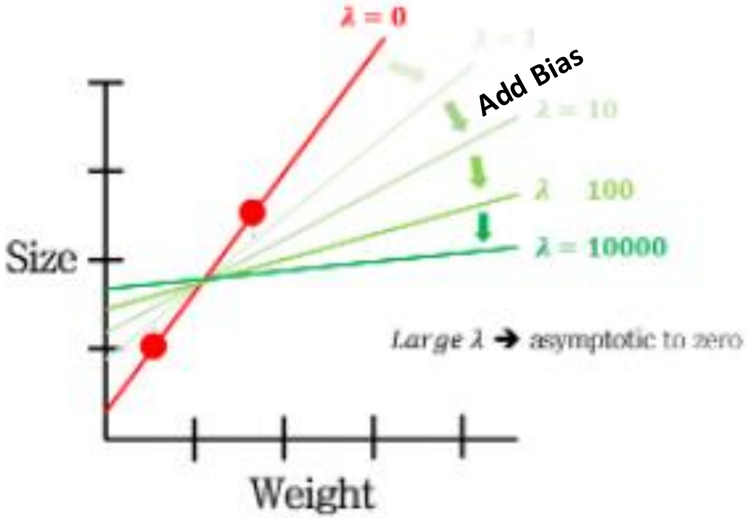
- Unlike C_p, AIC, and BIC, for which a small value indicates a model with a low-test error, a large value of adjusted R^2 indicates a model with a small test error.
- Maximizing the adjusted R^2 is equivalent to minimizing $\frac{SSR}{(n-d-1)}$. While SSR always decreases as the number of variables in the model increases, $\frac{SSR}{(n-d-1)}$ may increase or decrease, due to the **presence of d** in the denominator.
- The intuition behind the adjusted R^2 is that once all the correct variables have been included in the model, adding additional noise variables leads to an increase in d which will lead to increase in $\frac{SSR}{(n-d-1)}$ and consequently a decrease in adjusted R^2
- Unlike the R^2 statistic, the adjusted R^2 statistic pays a price for the inclusion of unnecessary variables in the model.

Regularization: Shrinkage with Ridge & Lasso









Ref: https://tyami.github.io/machine%20learning/regularization-Ridge-Lasso-ElasticNet/#google_vignette

Shrinkage Methods: Ridge Regression (L2)

- In addition to the subset techniques for selecting 'best' predictors, there are additional techniques which focuses on adding small amount of bias to the Ordinary least squares such that the model coefficients estimates can be 'shrunk' towards zero to reduce the variance in the estimates.
- Sum of Squares Residuals (SSR) measures how well a linear regression model matches training data. It is represented as:

$$\text{SSR} = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 .$$

- Ridge regression is written as:

$$\text{SSR} = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 . + \lambda \sum_{j=1}^{p} \beta_j^2$$

Where:
λ: hyper parameter tuned over various values

Ridge Regression

SSR + λ*sum of squared coefficients

$$\text{SSR} = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2$$

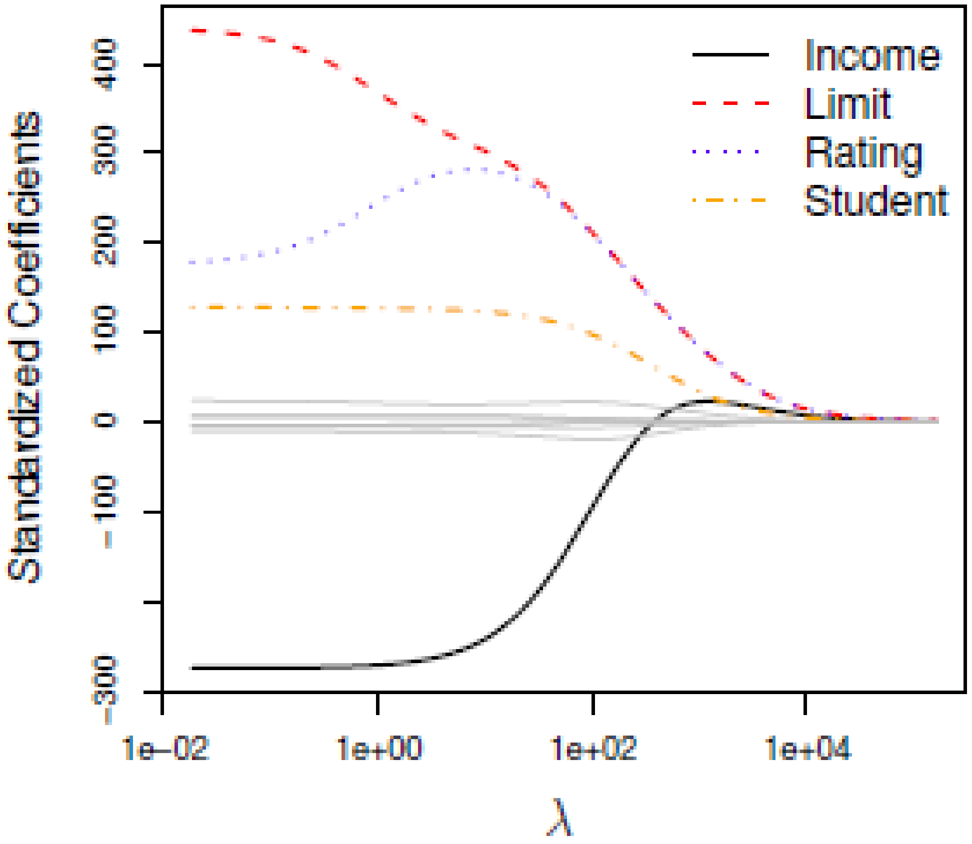
Some books lists this as $(\text{Slope})^2$

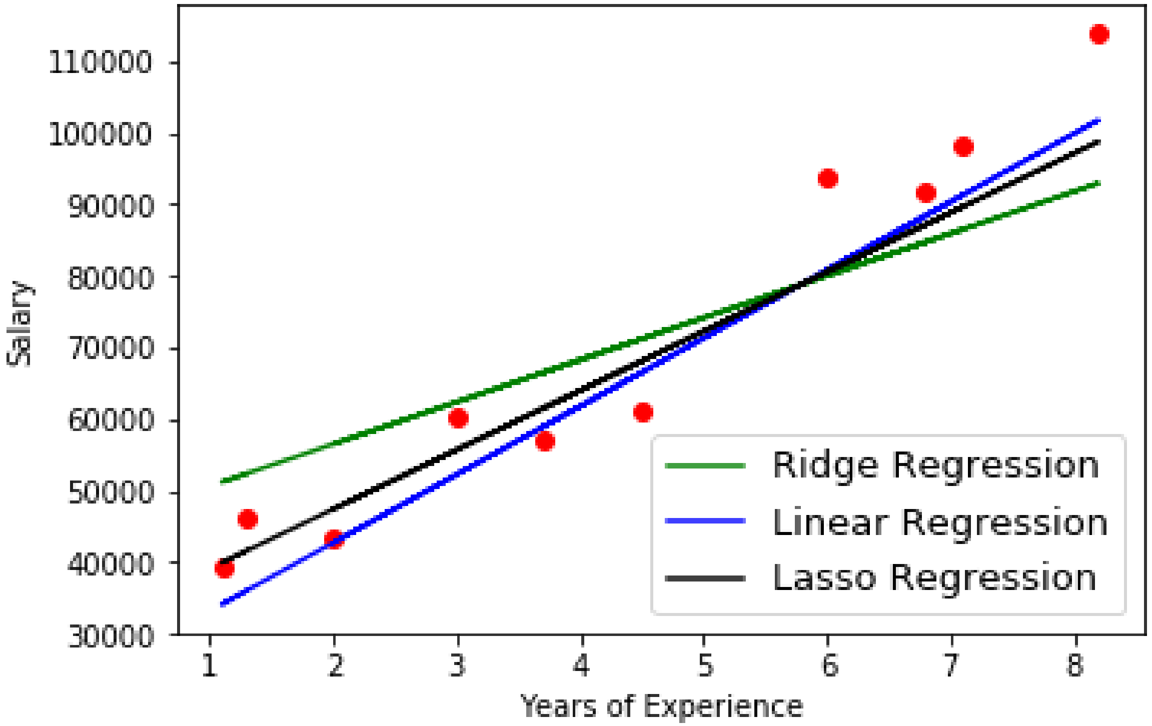
Hyper-parameter tuned over various values

Individual Betas of all predictors

- As with least squares, ridge regression seeks coefficient estimates that fits the data well, by making the SSR small.
- However, the second term called a shrinkage penalty, is small when $\beta_1, \ldots, \beta_p$ are close to zero, and so it has the effect of shrinking the estimates of **j** towards zero.
- The tuning parameter **λ** controls the strength of the constraint on the coefficients and acts as a tuning parameter. Selecting a good value for **λ** is critical; cross-validation is used for this.
- Note that ridge regression does not shrink every coefficient by the same value. Rather, coefficients are shrunk in proportion to their initial size. As **λ** increases, high-value coefficients shrink at a greater rate than low-value coefficients. High-value coefficients are thus penalized greater than low-value coefficients.

Ridge Regression





Note that the L2 penalty shrinks coefficients towards zero but never to absolute zero; although model feature weights may become negligibly small, they never equal zero in ridge regression.

With addition of small **λ,** the amount of change in dependent variable reduces for corresponding change in predictor values – as seen from comparing Linear & Ridge regression lines. This is interpreted as addition of small bias to make big reduction in variance

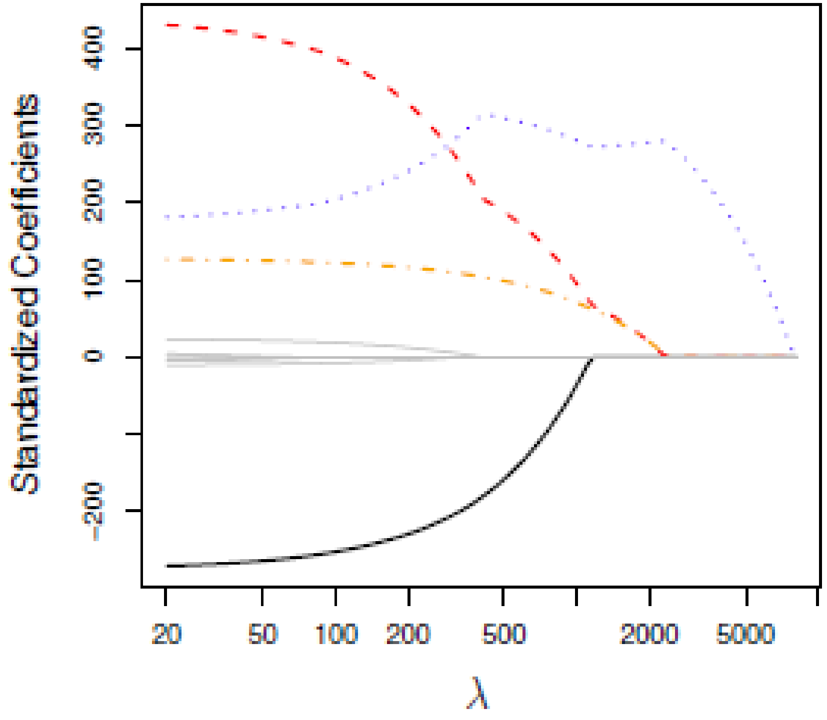
Lasso Regression (L1)

- Just as Ridge, Lasso is a regularization technique that applies a penalty to prevent overfitting and enhance accuracy of statistical models
- Ridge regression does have one obvious disadvantage, unlike subset selection, which will generally select models that involve just a subset of the variables, ridge regression will include all ***p*** predictors in the final model

$$\sum_{i=1}^{n}\left(y_i-\beta_0-\sum_{j=1}^{p}\beta_j x_{ij}\right)^2+\lambda\sum_{j=1}^{p}|\beta_j| = \text{SSR} + \lambda\sum_{j=1}^{p}|\beta_j|.$$

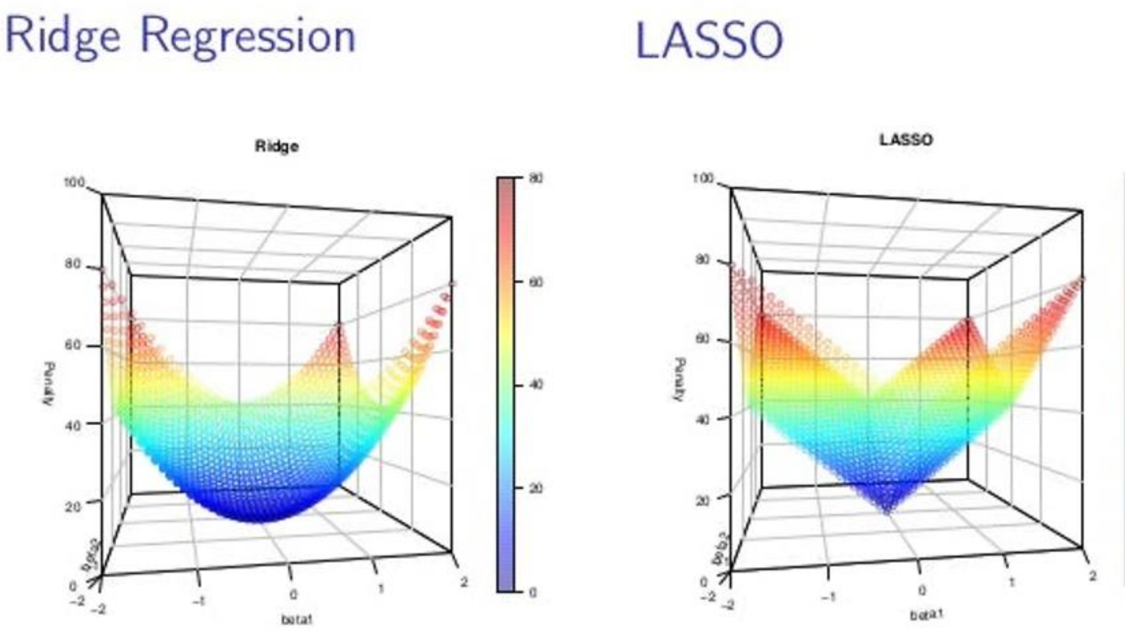
SSR + λ*sum of absolute coefficients

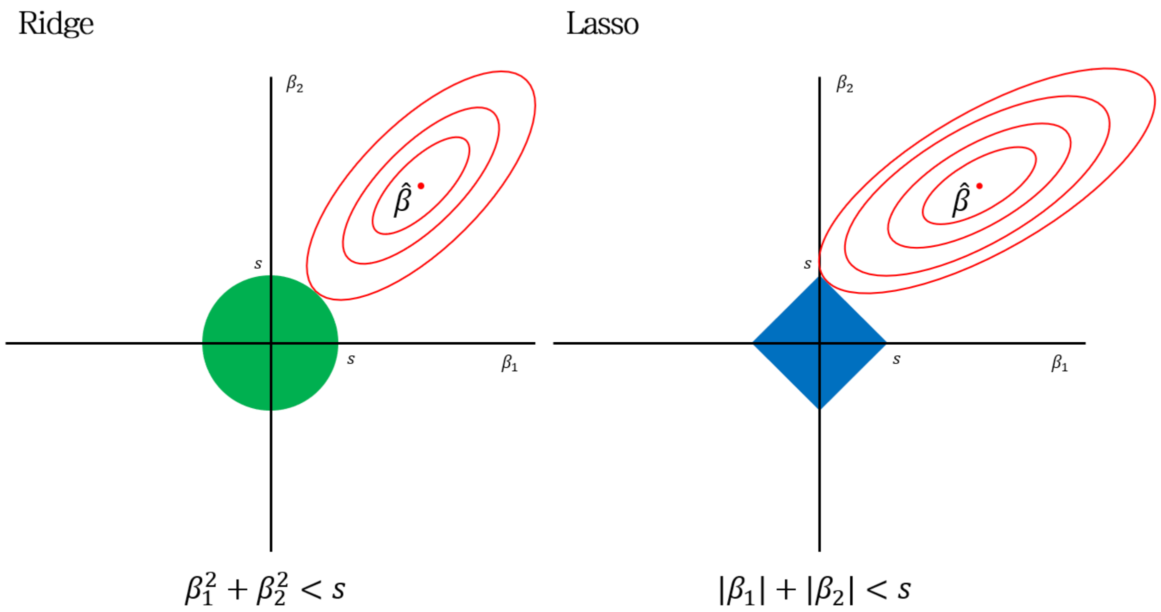
The Lasso



- As with ridge regression, the lasso shrinks the coefficient estimates towards zero.
- However, in the case of the lasso, the ℓ_1 penalty has the effect of forcing some of the coefficient estimates to be exactly equal to zero when the tuning parameter λ is sufficiently large.
- Hence, much like best subset selection, the lasso performs *variable selection*.
- We say that the lasso yields *sparse* models — that is, models that involve only a subset of the variables.
- As in ridge regression, selecting a good value of λ for the lasso is critical; cross-validation is again the method of choice.

Shrinkage Methods: 3D & 2D views





Ref: https://rishika-aditya0411.medium.com/l1-and-l2-regularization-lasso-and-ridge-regression-e9ef76e44c4e

Ref: https://tyami.github.io/machine%20learning/regularization-Ridge-Lasso-ElasticNet/#google_vignette

Conclusion

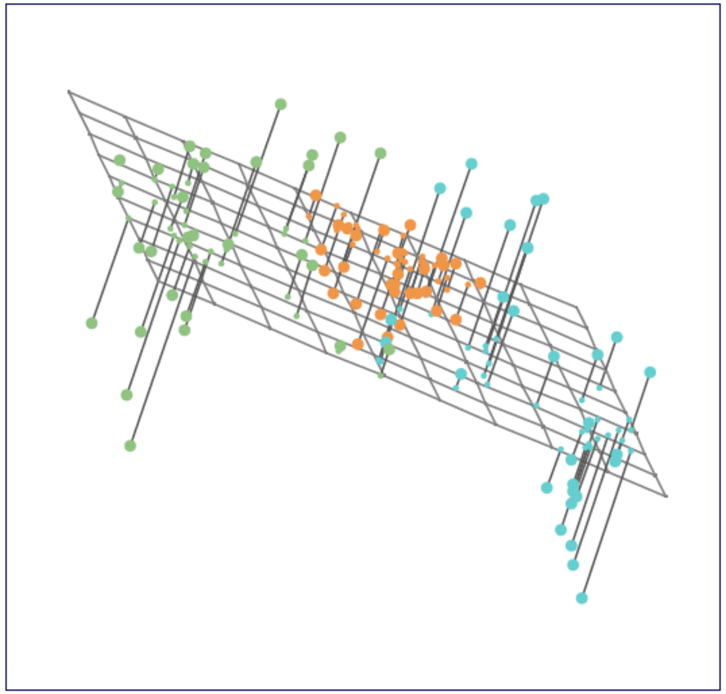
- These two examples illustrate that neither ridge regression nor the lasso will universally dominate the other.
- In general, one might expect the lasso to perform better when the response is a function of only a relatively small number of predictors.
- However, the number of predictors that is related to the response is never known *a priori* for real data sets.
- A technique such as cross-validation can be used in order to determine which approach is better on a particular data set.

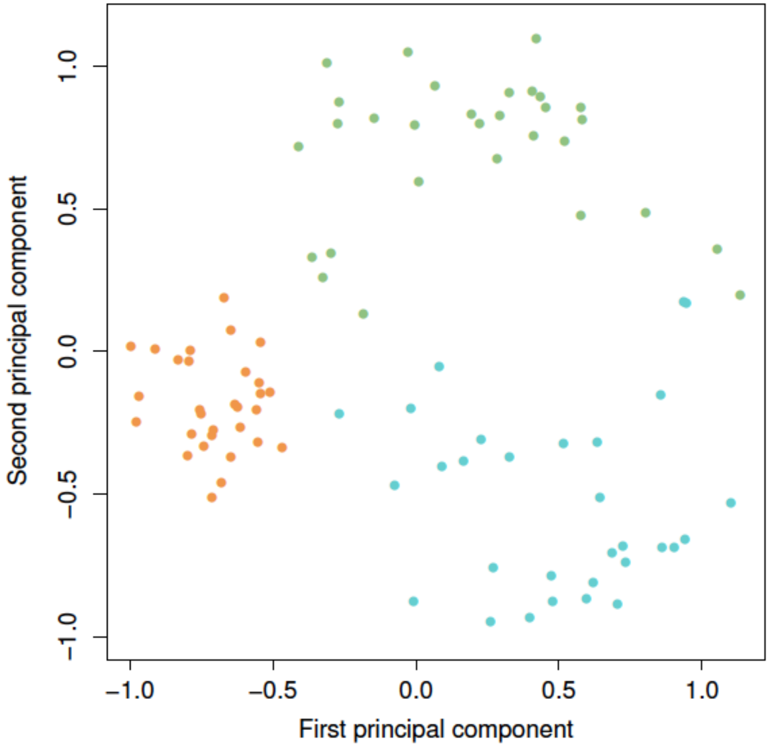
Selecting the tuning parameter for Ridge & Lasso

- As for subset selection, for ridge regression and lasso we require a method to determine which of the models under consideration is best.
- That is, we require a method selecting a value for the tuning parameter λ or equivalently, the value of the constraint s.
- *Cross-validation* provides a simple way to tackle this problem. We choose a grid of λ values, and compute the cross-validation error rate for each value of λ.
- We then select the tuning parameter value for which the cross-validation error is smallest.
- Finally, the model is re-fit using all of the available observations and the selected value of the tuning parameter.

Dimensionality Reduction using PCA

Principal Component Analysis





- Principal Component Analysis (PCA) is a statistical procedure that uses an orthogonal transformation for extracting information from a high-dimensional space by projecting it into a new coordinate system or lower-dimensional sub-space.
- Principal components have both direction and magnitude. The direction represents across which *principal axes* the data is mostly spread out or has most variance and the magnitude signifies the amount of variance that Principal Component captures of the data when projected onto that axis.
- It takes set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables called principal components.

Principal Component Analysis

- In this new system, the first axis corresponds to the first principal component, which is the component that explains the greatest amount of variance in the data.
- The second axis corresponds to the second principal component, which is orthogonal to the first and accounts for the next highest amount of variance, and so on. This process continues until as many components as the dimensionality of the data have been calculated. The result is a set of new variables, the principal components, which are uncorrelated and ordered in terms of the amount of variance they explain in the data.
- The reason you achieve uncorrelated principal components from the original features is that the correlated features contribute to the same principal component, thereby reducing the original data features into uncorrelated principal components; each representing a different set of correlated features with different amounts of variation.

Steps to calculate Principal Component Analysis

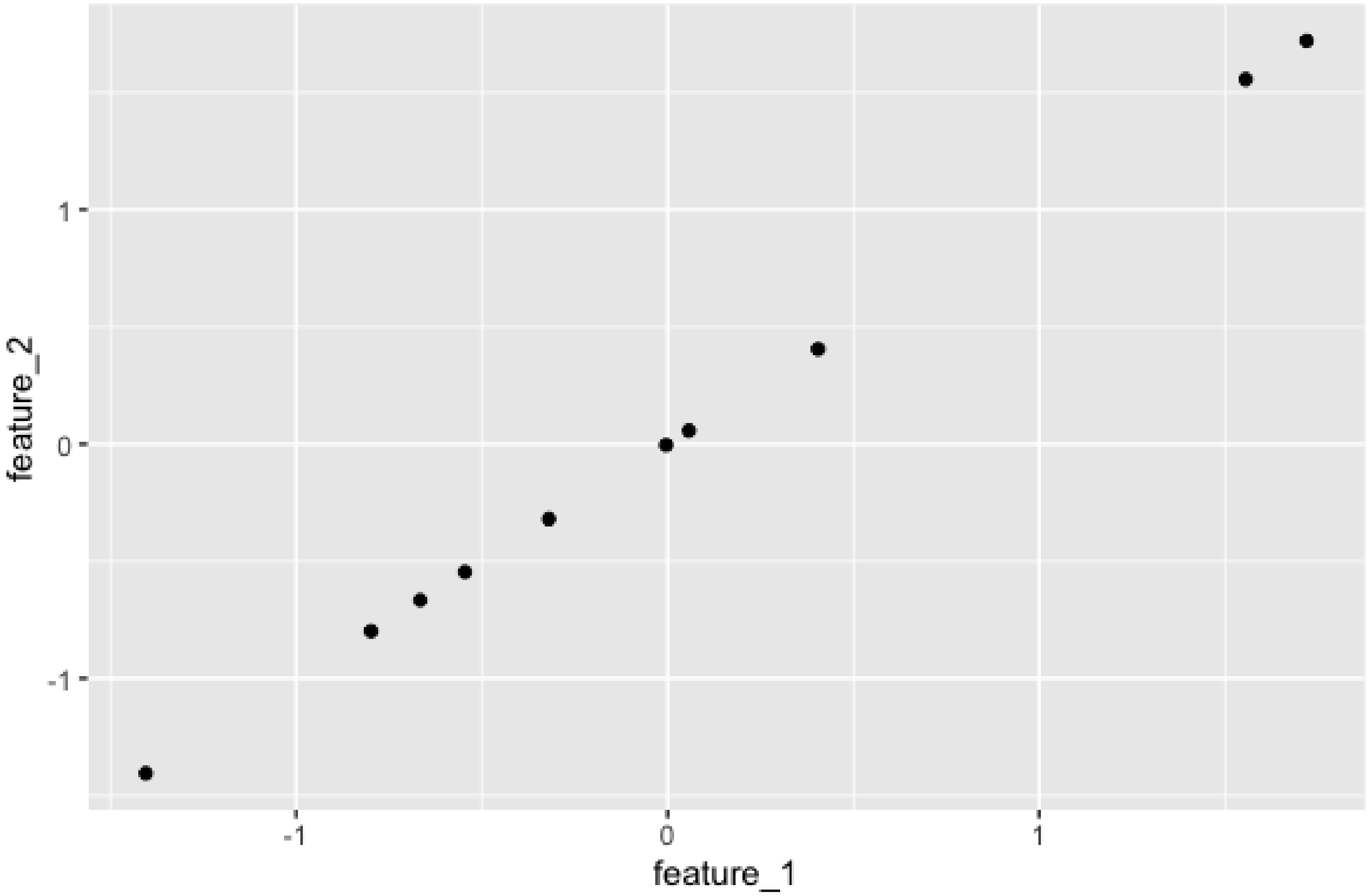
- Center the variables (some books also recommend to standardize)
- Compute the covariance matrix (identify correlations)
  - Measures strength of correlation between two variables
  - This is a square matrix i.e. for 3-dim data, it's a 3x3 matrix or 9 variable combinations
- Calculate Eigenvalues & Eigenvectors of the covariance matrix
  - Eigenvalues (λ) represent the amount of variance in each component
  - Eigenvectors (v) are principal components associated with Eigenvalues. Each Eigenvector points to a specific direction and their direction represents that direction of maximum variance.
  - Eigenvectors are typically normalized, meaning their length is 1

Principal Component Analysis

- What does it mean to ”find a low-dimensional representation of the data”?

feature_1	feature_2
-0.67	-0.67
-0.32	-0.32
1.56	1.56
0.0	0.0
0.06	0.06
1.72	1.72
0.41	0.41
-1.4	-1.4
-0.8	-0.8
-0.55	-0.55

Observations



Principal Component Analysis

- The observations fall along a one-dimensional line

feature_1	feature_2
-0.67	-0.67
-0.32	-0.32
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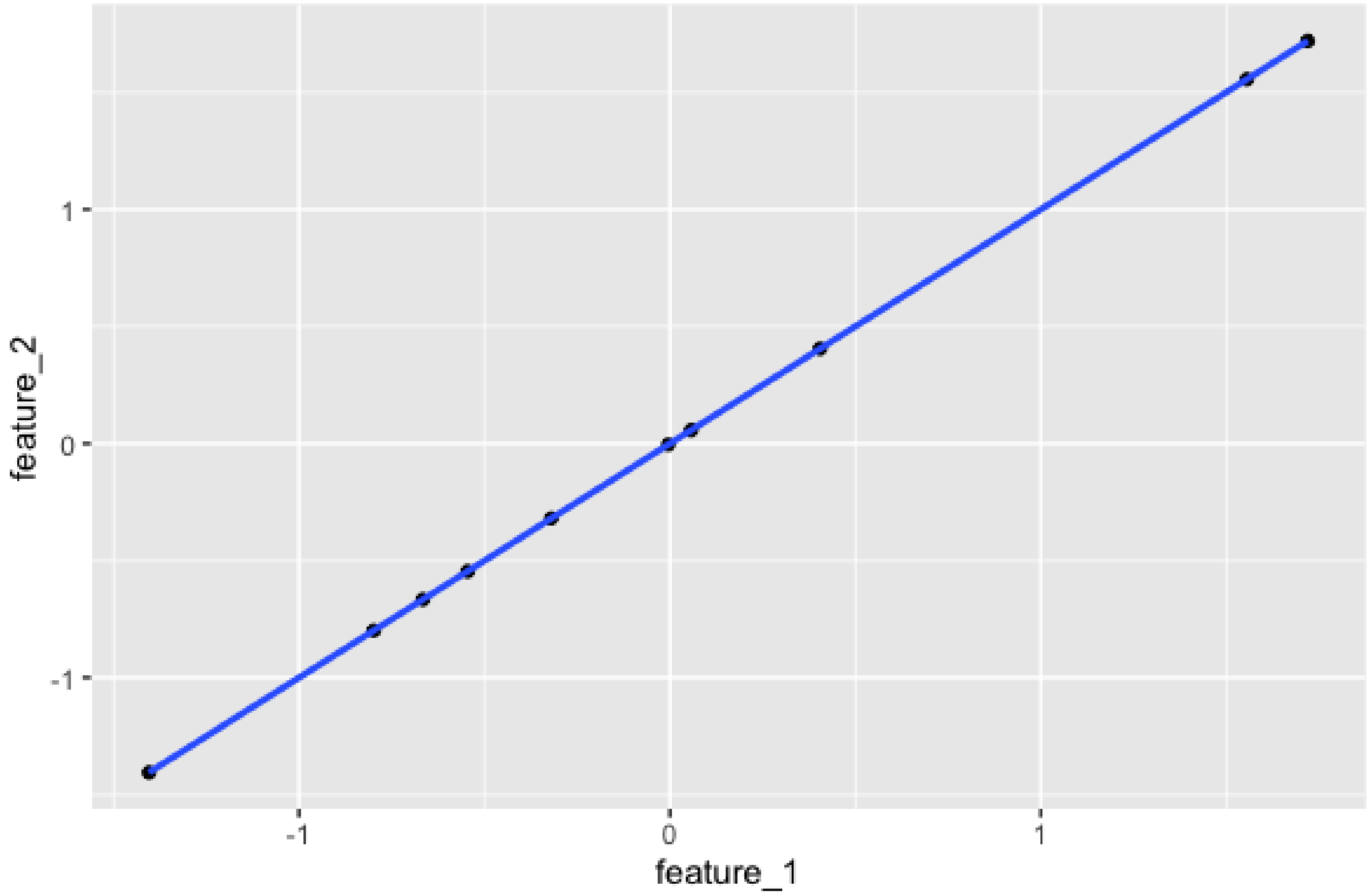

Principal Component Analysis

- Write the observations in terms of the unit vector in the direction of the one-dimensional line
  - The unit vector is called a principal component (PC)

feature_1	feature_2
-0.67	-0.67
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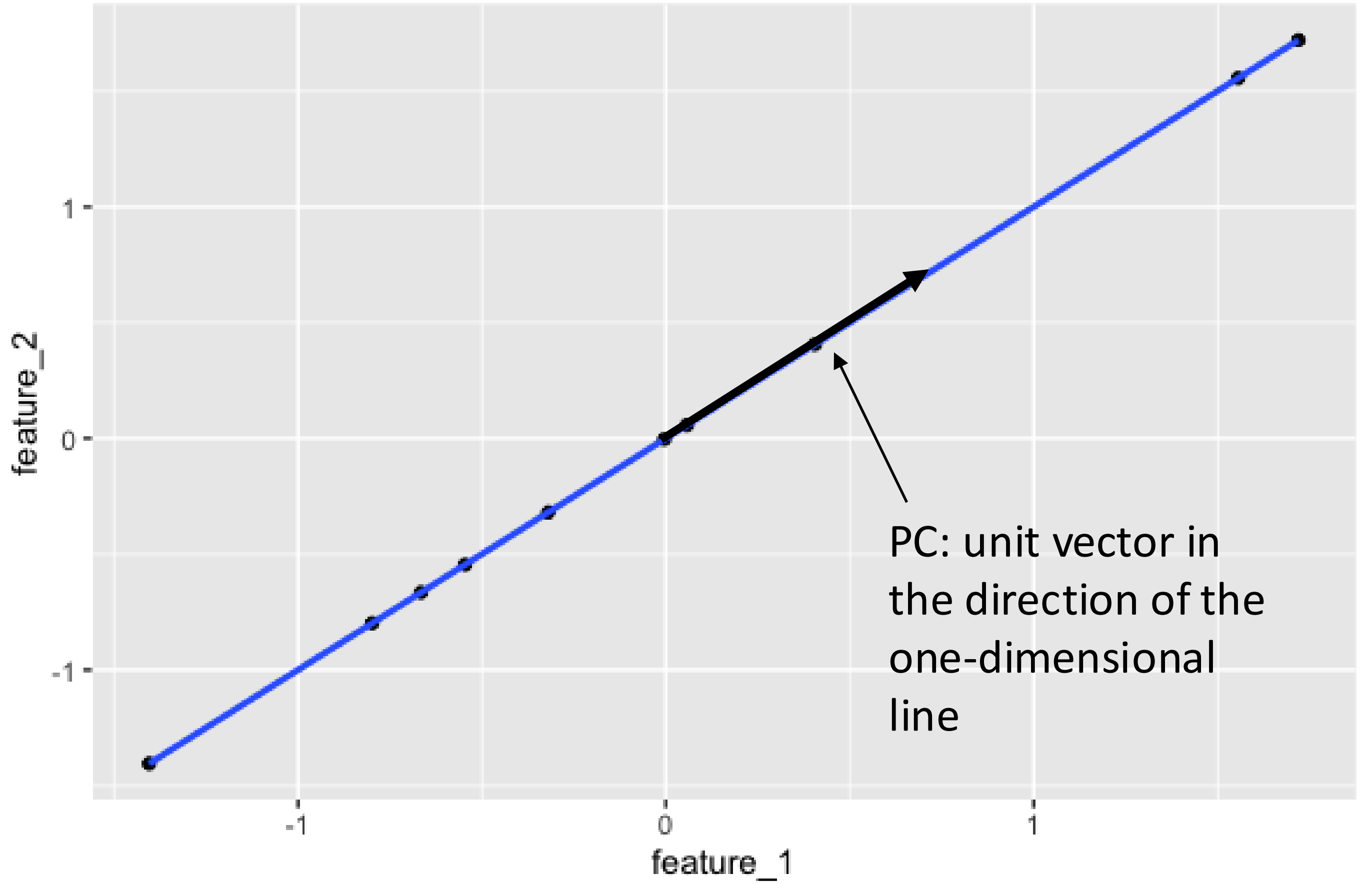
Observations

PC

Observations in terms of PC



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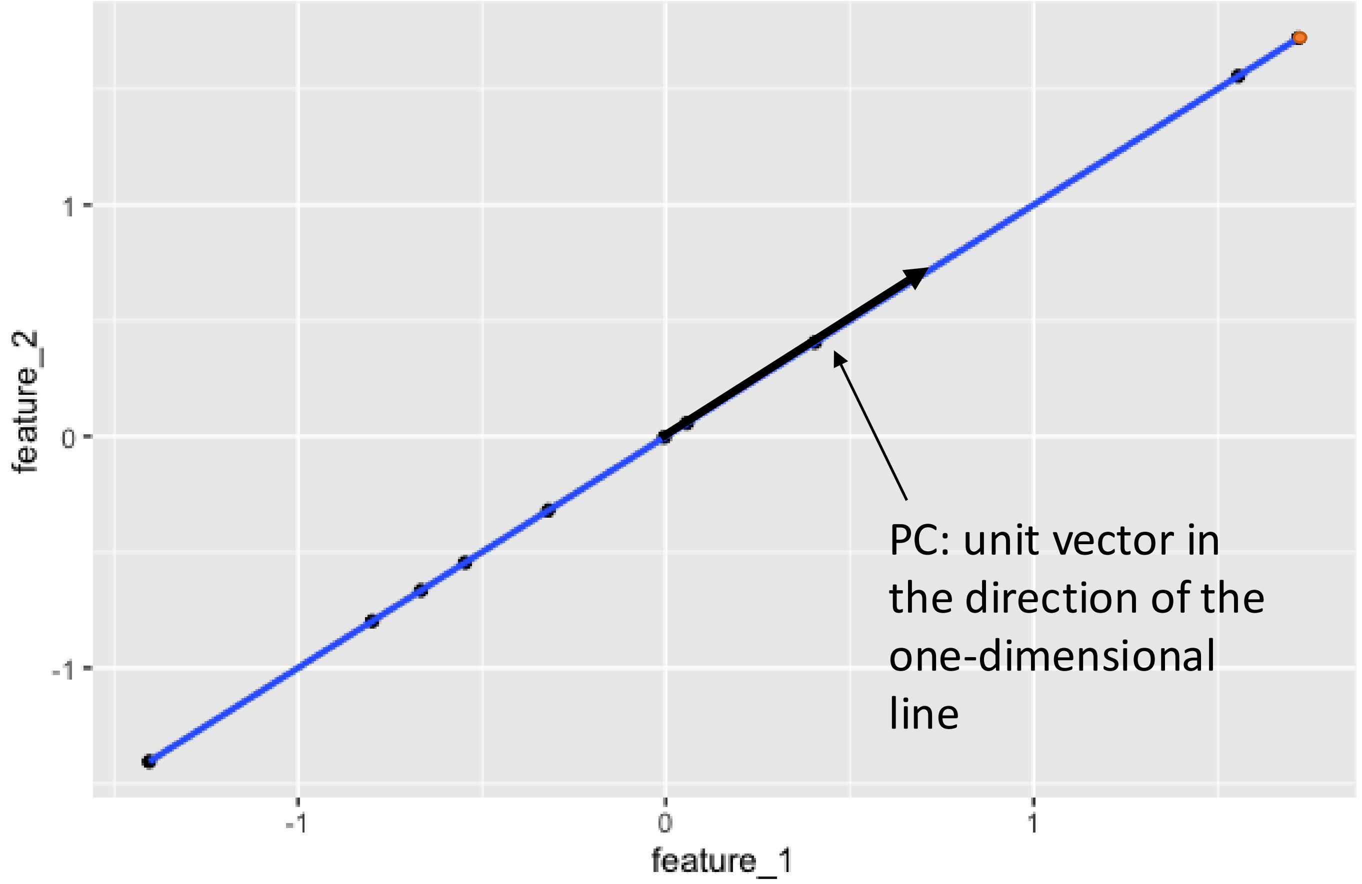
Observations

PC

Observations in terms of PC



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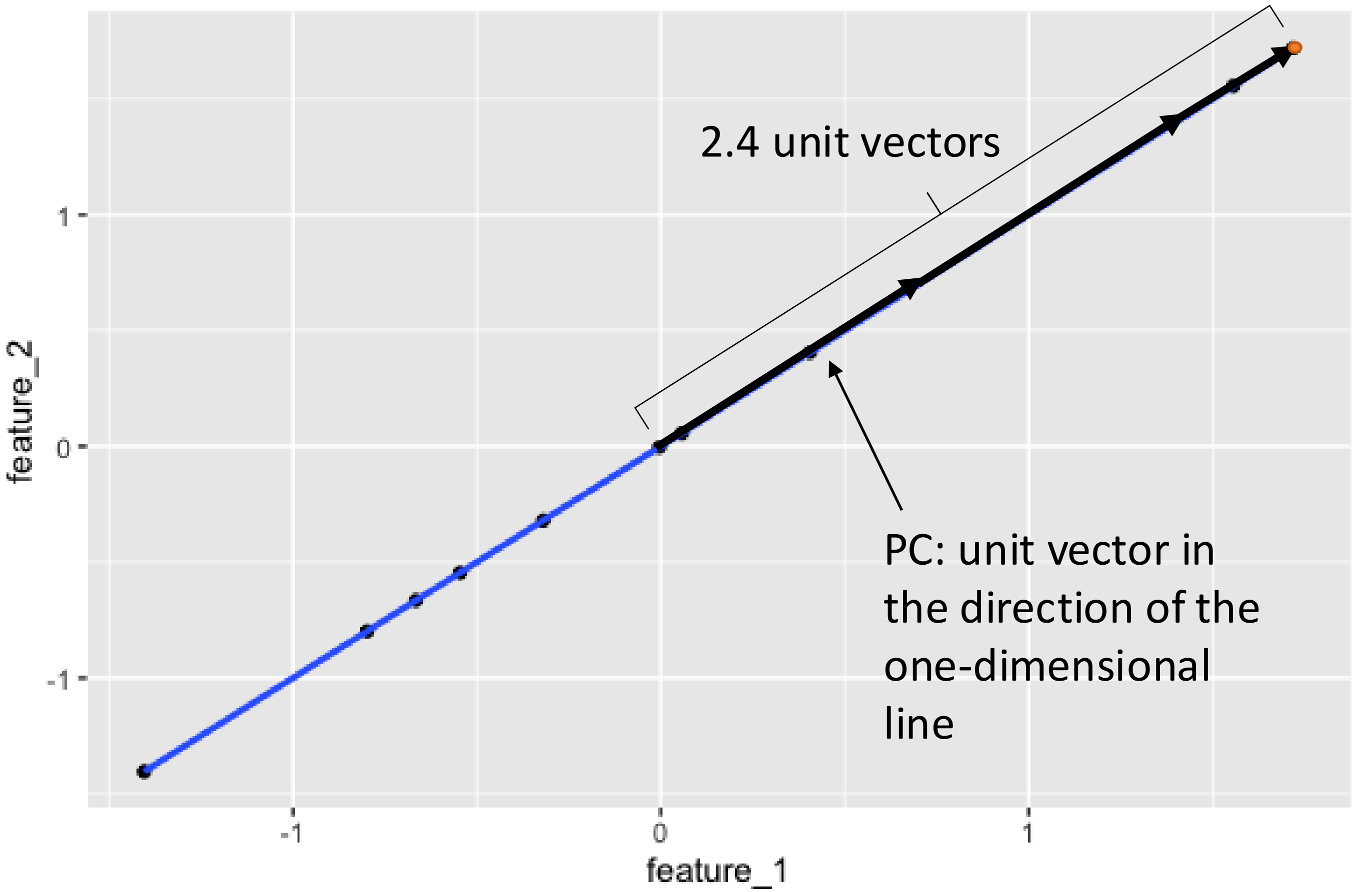
-0.8	-0.8
-0.55	-0.55

Observations

PC
2.4

Observations in terms of PC



Principal Component Analysis

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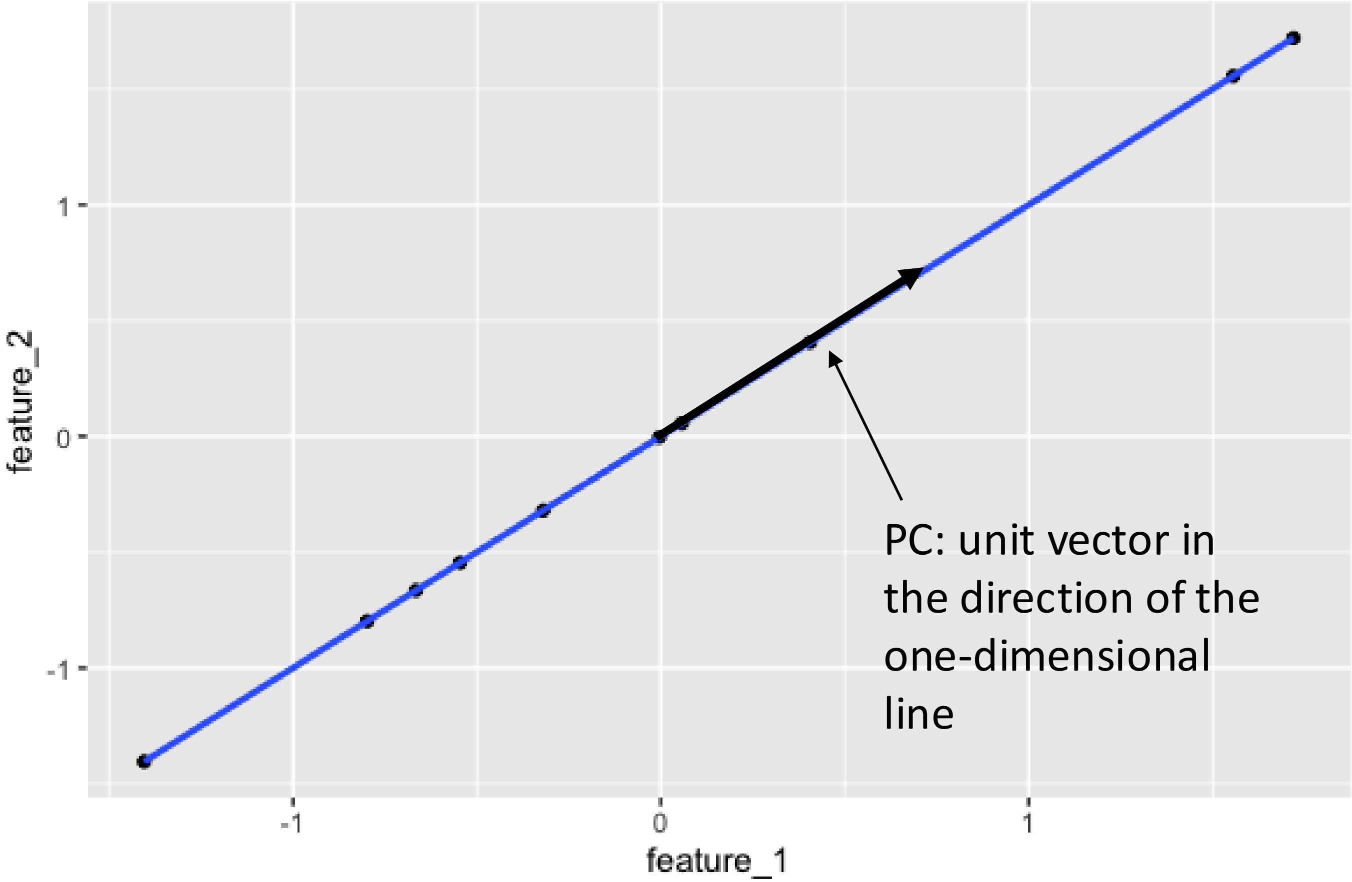
1.72	1.72
0.41	0.41
-1.4	-1.4
-0.8	-0.8
-0.55	-0.55

Observations

→

PC
-0.9
-0.5
2.2
0.0
0.1
2.4
0.6
-2.0
-1.1
-0.8

Observations in terms of PC

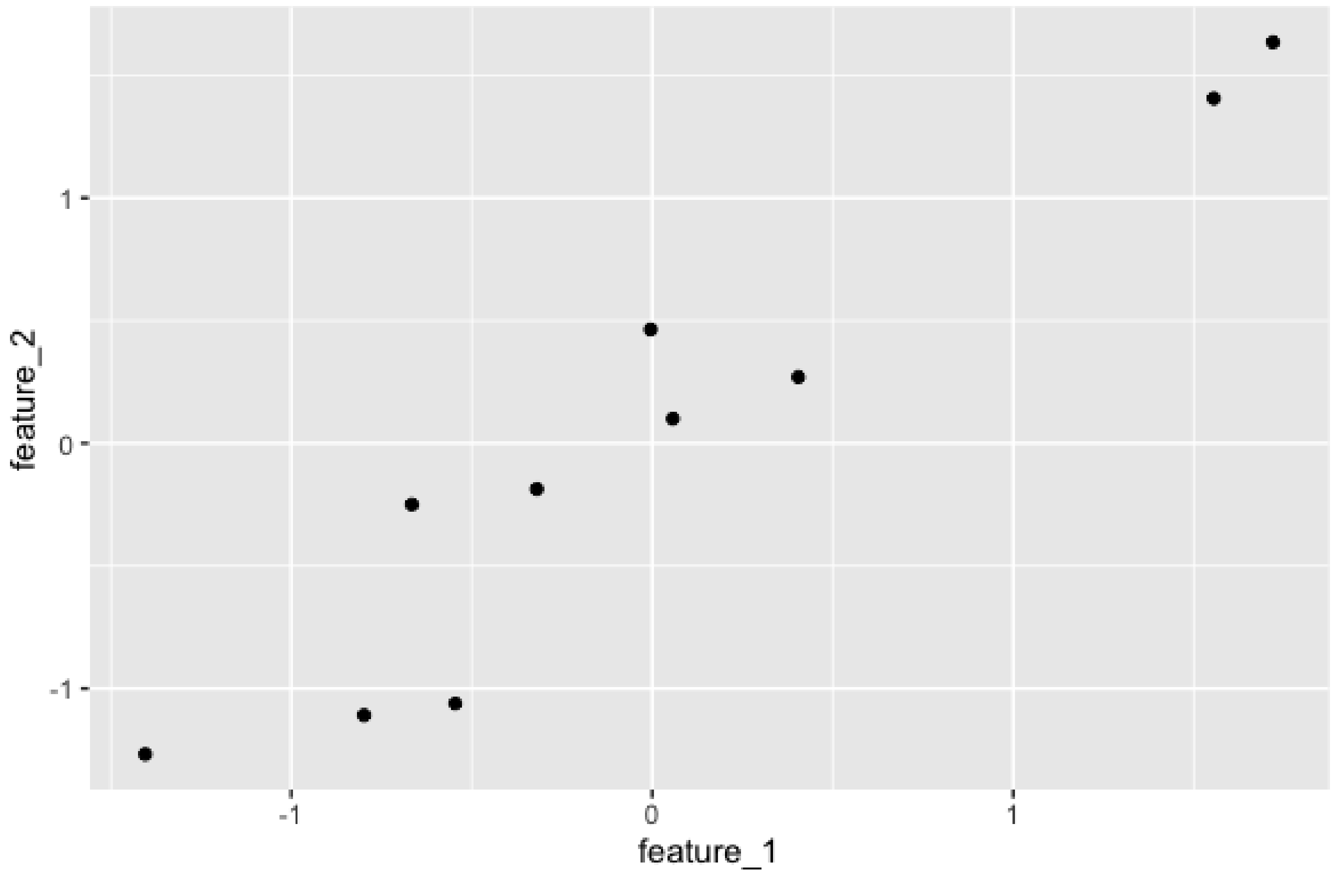


Principal Component Analysis

- What if the observations don't necessarily fall along a one-dimensional line, but are close?

feature_1	feature_2
-0.67	-0.25
-0.32	-0.19
1.56	1.41
0.0	0.46
0.06	0.1
1.72	1.63
0.41	0.27
-1.4	-1.27
-0.8	-1.11
-0.55	-1.06

Observations



Principal Component Analysis

- PC1 is the unit vector in the direction the data varies most; PC2 is the unit vector orthogonal to PC1
- Write the observations in terms of PC1 and PC2
  - PC1 is the best one-dimensional approximation to the observations → try using only PC1 to represent the data

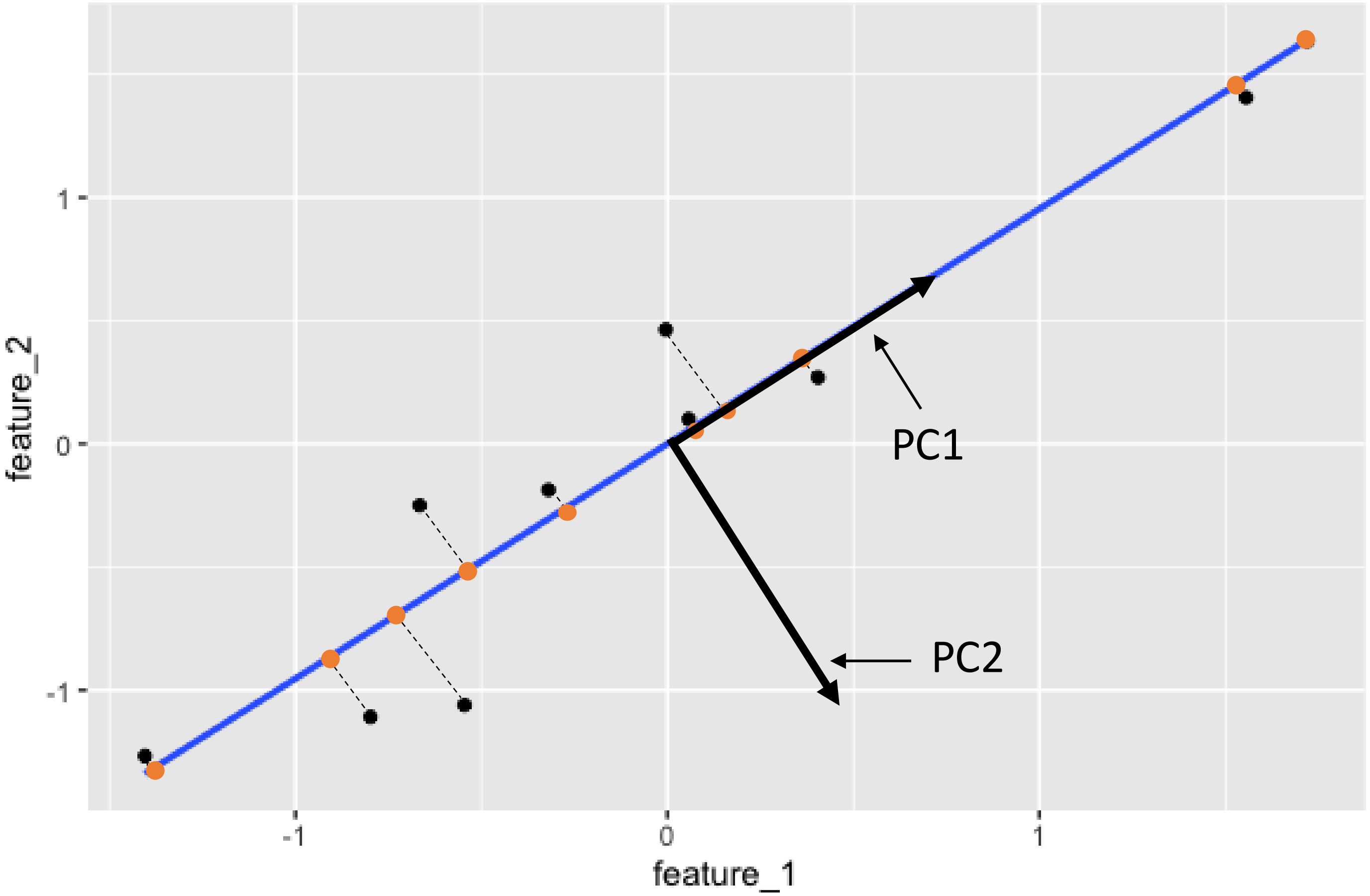
feature_1	feature_2
-0.67	-0.25
-0.32	-0.19
1.56	1.41
0.0	0.46
0.06	0.1
1.72	1.63
0.41	0.27
-1.4	-1.27
-0.8	-1.11
-0.55	-1.06

Observations

→

PC1	PC2
-0.65	-0.29
-0.36	-0.09
2.09	0.11
0.32	-0.33
0.11	-0.03
2.37	0.06
0.48	0.1
-1.89	-0.1
-1.35	0.22
-1.14	0.36

Observations in terms of PC1 and PC2

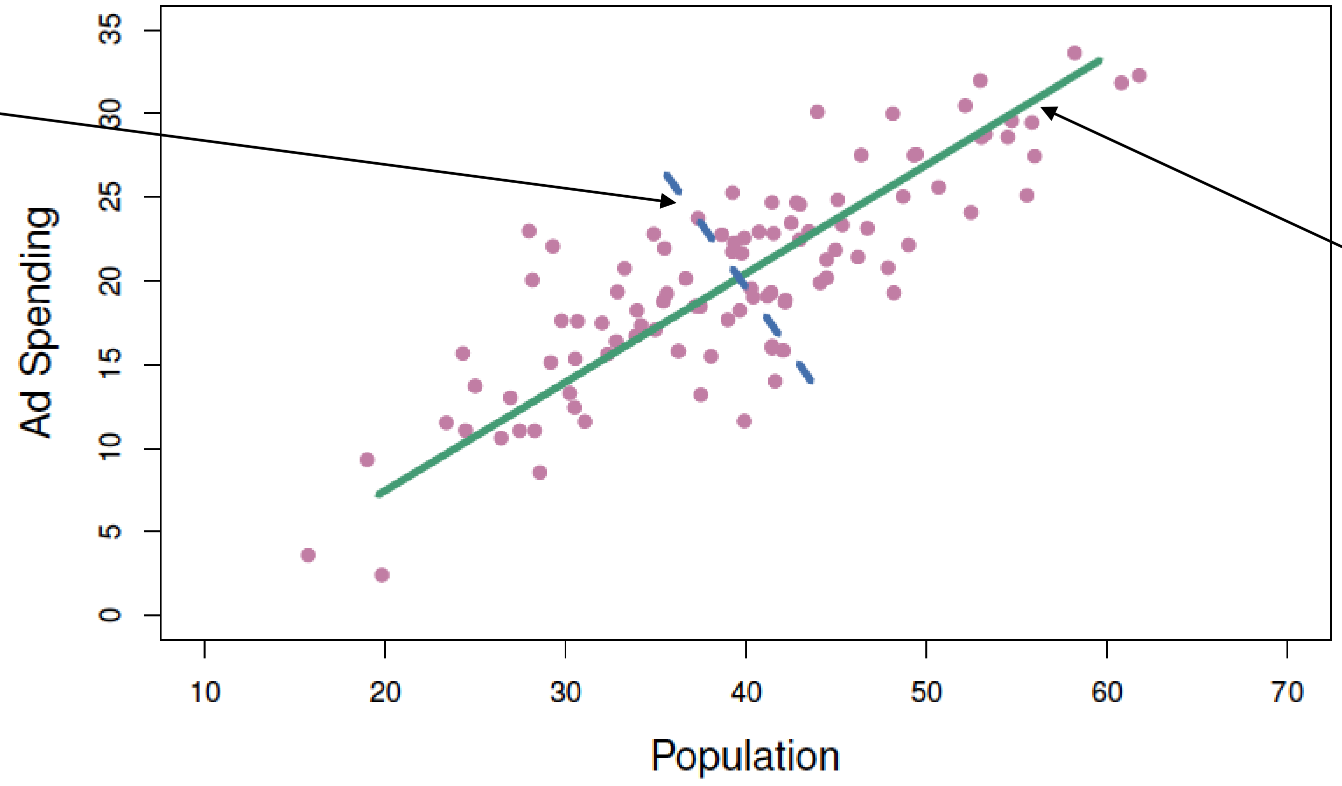


Principal Component Analysis

- Find the directions (unit vectors) in which the data varies most
  - Directions are found sequentially, with each direction being orthogonal to all previous directions
  - Directions are the PCs

Second direction, called the second PC.

If we project the data onto this direction, it has the largest variance out of all possible directions that are orthogonal to the first direction



First direction, called the first PC.

If we project the data onto this direction, it has the largest variance out of all possible directions

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2017) *An Introduction to Statistical Learning with Applications in R*. Springer